

Bulan 4: Real-Time & Async Features

Minggu 1: Django Channels & WebSockets

Hari 1: Channels Introduction & Setup

Project: RealtimeStarter

- Deskripsi: Channels boilerplate dengan ASGI configuration
- Fitur: WebSocket consumers, routing, Redis layer
- Keunikan: Health check dashboard untuk WebSocket connections

Hari 2: WebSocket Consumers

Project: ChatRoom

- Deskripsi: Real-time chat room dengan typing indicators
- Fitur: Multiple rooms, online users, message history
- Keunikan: Message reactions dan thread replies

Hari 3: Channel Layers & Groups

Project: LiveNotify

- Deskripsi: Notification system dengan real-time delivery
- Fitur: User-specific notifications, read/unread status
- Keunikan: Notification grouping dan smart batching

Hari 4: Database Integration with Channels

Project: CollabEditor

- Deskripsi: Collaborative document editor dengan live cursors
- Fitur: Multi-user editing, conflict resolution, version history
- Keunikan: Operational transform untuk concurrent edits

Hari 5: Authentication in WebSockets

Project: SecureSocket

- Deskripsi: WebSocket authentication dengan JWT tokens

- Fitur: Token-based auth, connection tracking, auto-reconnect
- Keunikan: Heartbeat mechanism untuk connection health

Hari 6: Broadcasting & Events

Project: LiveFeed

- Deskripsi: Activity feed dengan real-time updates
- Fitur: Follow system, like/comment notifications
- Keunikan: Smart feed algorithm dengan relevance scoring

Hari 7: Chat Application Complete

Project: ChatPro

- Deskripsi: Production-ready chat dengan advanced features
- Fitur: Direct messages, groups, file sharing, voice notes
- Keunikan: End-to-end encryption simulation, message search

Minggu 2: Celery & Background Tasks

Hari 8: Celery Setup & Configuration

Project: TaskQueue

- Deskripsi: Celery configuration dengan RabbitMQ/Redis
- Fitur: Task routing, priority queues, monitoring
- Keunikan: Dynamic queue creation based on load

Hari 9: Periodic Tasks & Cron

Project: SchedulePro

- Deskripsi: Scheduled task manager dengan crontab UI
- Fitur: Recurring jobs, task history, failure retry
- Keunikan: Natural language schedule input

Hari 10: Task Chains & Groups

Project: WorkflowEngine

- Deskripsi: Complex workflow orchestration dengan task dependencies

- Fitur: Sequential/parallel tasks, conditional execution
- Keunikan: Visual workflow builder dengan DAG representation

Hari 11: Email Sending Tasks

Project: MailQueue

- Deskripsi: Bulk email sender dengan template engine
- Fitur: Email campaigns, tracking, A/B testing
- Keunikan: Send time optimization based on timezone

Hari 12: Image Processing Tasks

Project: ImagePipeline

- Deskripsi: Image processing dengan thumbnail generation
- Fitur: Resize, crop, filters, watermark, format conversion
- Keunikan: AI-powered auto-crop untuk faces

Hari 13: Data Export & Import Tasks

Project: DataExchange

- Deskripsi: Large dataset import/export dengan progress tracking
- Fitur: CSV, Excel, JSON formats, validation, scheduling
- Keunikan: Incremental import dengan change detection

Hari 14: Task Monitoring & Error Handling

Project: CeleryWatch

- Deskripsi: Celery monitoring dashboard dengan Flower alternative
- Fitur: Task stats, worker health, error logs, retry management
- Keunikan: Predictive scaling recommendations

Minggu 3: Advanced Frontend Integration

Hari 15: HTMX Integration

Project: HTMXDash

- Deskripsi: Dynamic UI dengan HTMX (no JavaScript framework)
- Fitur: Infinite scroll, live search, modal forms
- Keunikan: Server-side rendering dengan SPA-like UX

Hari 16: Alpine.js with Django

Project: AlpineDjango

- Deskripsi: Interactive components dengan Alpine.js
- Fitur: Dropdown, tabs, accordions, tooltips
- Keunikan: Reusable component library

Hari 17: Django + React Setup

Project: ReactDjango

- Deskripsi: Django backend dengan React frontend (SPA)
- Fitur: JWT auth, API integration, routing
- Keunikan: Server-side rendering untuk SEO

Hari 18: State Management

Project: StateFlow

- Deskripsi: Complex state management dengan Redux/Context
- Fitur: Global state, async actions, persistence
- Keunikan: Time-travel debugging

Hari 19: Real-Time Dashboard

Project: MetricsDash

- Deskripsi: Real-time analytics dashboard dengan charts
- Fitur: Live data updates, multiple visualizations, filters
- Keunikan: Customizable widget layout (drag-drop)

Hari 20: File Upload with Progress

Project: UploadStation

- Deskripsi: Modern file uploader dengan chunking

- Fitur: Drag-drop, progress bars, pause/resume, preview
- Keunikan: Client-side image compression dan validation

Hari 21: Progressive Web App

Project: PWAKit

- Deskripsi: Django PWA dengan offline support
- Fitur: Service worker, manifest, push notifications
- Keunikan: Smart caching strategy untuk optimal performance

Minggu 4: Third-Party Integrations

Hari 22: Payment Integration (Stripe)

Project: PayFlow

- Deskripsi: Payment processing dengan Stripe integration
- Fitur: One-time payments, subscriptions, webhooks
- Keunikan: Multi-currency support dengan auto-conversion

Hari 23: Cloud Storage (S3/Cloudinary)

Project: CloudStore

- Deskripsi: File storage dengan CDN integration
- Fitur: Direct upload to S3, signed URLs, image optimization
- Keunikan: Automatic backup rotation policy

Hari 24: Social Media APIs

Project: SocialHub

- Deskripsi: Social media aggregator (Twitter, Instagram APIs)
- Fitur: Cross-posting, analytics, scheduling
- Keunikan: Unified content calendar

Hari 25: Maps & Geolocation

Project: MapQuest

- Deskripsi: Location-based app dengan Google Maps

- Fitur: Store locator, geocoding, distance calculation
- Keunikan: Route optimization untuk multiple destinations

Hari 26: Analytics & Tracking

Project: InsightTrack

- Deskripsi: Custom analytics dengan event tracking
- Fitur: Page views, events, funnels, cohorts
- Keunikan: Privacy-focused (no cookies) tracking

Hari 27: Search Integration (Elasticsearch)

Project: ElasticSearch

- Deskripsi: Advanced search dengan Elasticsearch
- Fitur: Full-text search, faceted search, autocomplete
- Keunikan: Typo tolerance dan synonym handling

Hari 28: SMS & Push Notifications

Project: NotifyAll

- Deskripsi: Multi-channel notification system
- Fitur: SMS (Twilio), push, email, in-app
- Keunikan: User preference management dan quiet hours

Hari 29-30: Integration Project

Project: EventMaster

- Deskripsi: Event management platform dengan ticketing
- Fitur: Event creation, ticket sales (Stripe), QR codes, check-in app, email reminders, maps, social sharing, analytics
- Keunikan: Virtual event support dengan WebRTC, waitlist automation
- Scope: 2 hari untuk full-featured event platform